

TIC 348884775

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 6.731631 d
Epoch (t0) 1519.7050 BTJD
Duration 0.1700 d
Depth 14.751 % (147,512 ppm)
SNR 111.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 80.0 (gate ≥ 7.0)
Odd/even z-score 0.39 (consistent)
Secondary eclipse detected, phase 0.43, 7.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 8 light-curve dip in this sector: 66.5 sigma t0 refined by 0.4425 phase
Difference-image peak SNR 17.82 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 348884765
4.70 px away, dTmag +0.97 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

